Estimated enterolignans, lignan-rich foods, and fibre in relation to survival after postmenopausal breast cancer

Background: Lignans – oestrogenic substances present in various foods – are associated with postmenopausal breast cancer risk, but not much is known regarding their effects on survival. Methods: In a follow-up study of 2653 postmenopausal breast cancer patients diagnosed between 2001 and 2005, vital status and causes of death were verified through end of 2009. Hazard ratios (HRs) and 95% confidence intervals (CIs) for estimated enterolignans, lignan-rich foods, and dietary fibre in relation to overall survival (OS) and breast cancer-specific survival (BCSS) were assessed using Cox proportional hazards models stratified by age at diagnosis and adjusted for prognostic/confounding factors. Results: Median follow-up time was 6.4 years, and 321 women died, 235 with breast cancer. High estimated enterolactone and enterodiol levels were associated with significantly lower overall mortality (highest quintile, HR=0.60, 95% CI=0.40–0.89, PTrend=0.02 and HR=0.63, 95% CI=0.42–0.95, PTrend=0.02, respectively). Fibre intake was also associated with a significantly lower overall mortality. Differentiated by median fibre intake, associations with estimated enterolignans were still evident at low but not high fibre intake. There was no effect modification by oestrogen receptor status and menopausal hormone therapy. Conclusion: Postmenopausal breast cancer patients with high estimated enterolignans may have a better survival.